Nathan Riek

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SUMMARY

PhD-trained machine learning engineer with 5+ years building scalable data pipelines, machine learning models, and user-friendly applications, with deep experience in physiological timeseries data.

EXPERIENCE

Founding Machine Learning Engineer | IntelliCardio | Remote | Sep 2025 – Present

- Built an LLM-powered chatbot that answers ECG-related questions using LangChain and a RAG pipeline, ensuring medically grounded responses with inline source citations (deployed on Streamlit Community Cloud)
- Built a desktop application that plots ECGs, deidentifies ECG files, and generates median beats, features, and machine learning model predictions for non-technical users

Postdoctoral Research Associate | University of Rochester | Rochester, NY | Sep 2025 – Present

- Deploying a machine learning–driven decision support tool within a hospital Azure cloud environment
- Modified ECG preprocessing pipeline from PhD research for compatibility with the AliveCor KardiaMobile 6L at-home device

Data Science Intern | HeartBeam | Remote | Sep 2024 – Dec 2024

- Trained deep learning models in PyTorch to estimate cardiac age from noisy ECG time-series data
- Automated data preparation and prediction retrieval using Selenium, minimizing manual effort and facilitating scalable analysis of ECG datasets

PhD Researcher | University of Pittsburgh | Pittsburgh, PA | Jan 2021 – Aug 2025

- Designed ECG and EEG preprocessing pipelines using Python and MATLAB, adopted by 5+ research institutions
- Developed PyTorch models for ECG-based classification of occlusion myocardial infarction, winning the 2024 ISCE Early Career Investigator Award
- Built an explainable machine learning decision support platform in Dash and Azure to assist clinicians with ECG interpretation
- Published in top-tier journals including Nature Medicine, European Heart Journal, and IEEE Transactions on Biomedical Engineering

Engineering Co-Op | Eaton | United States | Sep 2018 – Aug 2020

- Built a C# GUI test automation tool, integrated RF instrumentation into TestStand/LabVIEW workflow, and contributed to Lean Six Sigma initiatives during my three Co-Op rotations

EDUCATION

PhD, Electrical and Computer Engineering
University of Pittsburgh | Jan 2021 – Aug 2025
BS, Electrical Engineering
University of Pittsburgh | Aug 2016 – Dec 2020

CERTIFICATIONS

IBM Data Science Professional Certificate | Coursera | Nov 2025
IBM Generative AI Engineering Professional Certificate | Coursera | Feb 2026

SKILLS

Machine Learning & Modeling: PyTorch, Scikit-learn
Programming: Python, R, MATLAB
Data Engineering & Automation: Pandas, NumPy, SciPy, SQL, Selenium, LangChain
Application Development: Dash, Streamlit, Gradio, PySide6
Visualization: Plotly, Matplotlib, Seaborn
Cloud & Deployment: Azure, AWS, Docker
Version Control: Git, GitHub